

UNITED STATES OF AMERICA
FEDERAL ENERGY REGULATORY COMMISSION

Interconnection of Large Loads to the Interstate
Transmission System

Docket No. RM26-4-000

COMMENTS OF THE UNION OF CONCERNED SCIENTISTS

The Union of Concerned Scientists (“UCS”) appreciates the opportunity to submit comments on Secretary of Energy’s advanced notice of proposed rulemaking (“proposed ANOPR”). UCS supports the Commission’s effort and offers these comments in Docket No. RM26-4-000 noticed October 27, 2025 regarding Interconnection of Large Loads to the Interstate Transmission System.

The Union of Concerned Scientists is a national nonprofit organization representing our members and the public interest in advancing responsible public policies in areas where science and technologies play a critical role. Established in 1969, UCS has created a unique alliance between many of the nation’s leading scientists and thousands of committed citizens. The UCS Climate & Energy Program focuses on developing a sustainable energy system—one that is affordable and nondepletable, and that does not degrade natural systems or public health. UCS is headquartered in Cambridge, Massachusetts and has offices in Oakland, Chicago and Washington, D.C.

We encourage the Commission to address the issues identified in the ANOPR and gather support and perspective for possible reforms that will provide the industry and consumers with increased reliability under changing conditions and protection from excessive costs. UCS submitted comments in a proceeding earlier this year related to

interconnection of large loads¹ as well as interconnection-related rulemakings over the past 12 years.²

I. Comments

UCS offers comments on Principles Five through Eight included in the proposed ANOPR.

A. Principle 5: Studies of large loads with on-site, grid connected generation must account for both the net and gross amounts of both load and generation because power limiting controls are not instantaneous

The fifth principle in the ANOPR suggests that “hybrid facilities” (defined as “large loads [...] seeking to share a point of interconnection with new or existing generation facilities”³) should be studied at their net capacity:

Fifth, hybrid facilities should be studied based on the amount of injection and/or withdrawal rights requested. For example, a hybrid facility consisting of a 500 MW load and a 600 MW generating facility may seek no withdrawal rights and 100 MW of injection rights. This provides incentives for co-location with new generation facilities and ensures efficient buildout of the transmission system.⁴

The example implies that rather than undergoing two separate study processes—one as a 500 MW load and one as a 600 MW generating facility—the project should only be studied as a 100 MW generating facility. While the intent to simplify the process and avoid the need to

¹ Order Instituting Proceeding Under Section 206 of the Federal Power Act and Consolidating With Other Proceedings Docket Nos. EL25-49-000, AD24-11-000, EL25-20-000 (Consolidated).

² See e.g. Small Generator Interconnection Agreements and Procedures RM13-2-000, Document Accession No. 20130531-5220; Essential Reliability Services and the Evolving Bulk-Power System - Primary Frequency Response RM16-6-000 Ascension No. 20170123-5154; Building for the Future Through Electric Regional Transmission Planning and Cost Allocation and Generator Interconnection RM21-17-000, Ascension Number 20211012-5493; Improvements to Generator Interconnection Procedures and Agreements RM22-14-000 Ascension No. 20221209-5169; Reliability Standards for Frequency and Voltage Protection Settings and Ride-Through For Inverter-Based Resources RM25-3-000, Ascension No. 20250207-5022.

³ Chris Wright, “Ensuring the Timely and Orderly Interconnection of Large Loads: Advance Notice of Proposed Rulemaking,” October 23, 2025, 9, <https://www.energy.gov/sites/default/files/2025-10/403%20Large%20Loads%20Letter.pdf>.

⁴ Ibid., 11.

perform two separate studies is reasonable, even a combined process must still consider both the full generation and full load amounts separately at specific points in the study process.

Considering only the net injection ignores the potential impacts of the project to the grid during normal operations as the moment-by-moment injection could vary widely, from -500 MW when the generator is offline up to +600 MW when the load is offline. While the sixth principle (discussed below) addresses controls and protections that will help mitigate unauthorized withdrawals and injections, technical limitations of the equipment in question (generators, load, and system protection facilities), still necessitate that the study process consider both the full withdrawal and the full injection of which the hybrid facility is capable. To put a finer point on it: capabilities to limit injections or withdrawals are not instantaneous and are not error-proof.

The challenge of meeting demand from new large loads requires the industry to recognize lessons learned from the widespread deployment of new generators on existing, often capacity-constrained distribution systems. Principles 5, 6, and 7 rely on concepts of controlling generation to abide by pre-determined limits regardless of variability in loads or inabilities of generator flexibility. Thermal generation operating on the bulk power system have generally not been required to adhere to such exacting performance expectations in the past because the size of most balancing areas and the scale of system operations was more forgiving of conventional plant inflexibility. The drafting and adoption of rules that presume tight control and a narrow range of performance from conventional generation with co-located loads will need to be based on clearly stated and explicitly measured standards or limits.

Similar considerations for inverter-based resources (“IBRs”) connected to the distribution system are instructive here. In many cases, the ability of multiple customers to connect distributed energy resources (“DERs”) to the same segment of the distribution system is

limited by the capacity of distribution equipment on that segment. To address this, many distribution utilities now allow the use of export-limiting systems. While this can be implemented in several ways, in recent years IBR manufacturers have begun offering inverters with built-in power control systems (“PCS”) that can limit export below specified kW limits. These inverters are now covered by several relevant standards, such as IEEE 1547-2018, UL 1741, and UL 3141.

However, there are two key caveats with this capability. First, power limiting is not instantaneous. When power output exceeds specified limits, a few moments are required for the PCS to measure this, then a few moments more to reduce the output. This phenomenon, known as inadvertent export, is an issue for both transmission and distribution systems, as discussed in a recent Department of Energy report on DER interconnection::

Inadvertent exports are minimal, short-duration power outputs from DERs, which occur during rapid changes in generation or load due to response delays from the plant’s PCS. For example, the interconnection agreement for a 750-kW PV system might allow the system to export no more than 500 kW to the grid, so any export above 500 kW would be inadvertent. Inadvertent export could have adverse voltage, thermal, protection, or power-quality impacts on the system.⁵

The report notes that inadvertent export events can be triggered by changes in generation or load. This is because just like the “hybrid facilities” discussed in the ANOPR, distribution-connected inverters almost universally share a point of common coupling with a load, and the export limit is net of load and generation—if either changes suddenly (for example, due to an air

⁵ Diane Baldwin et al., *Distributed Energy Resource Interconnection Roadmap; Transforming Interconnection by 2035*, Interconnection Innovation E-Xchange (I2x) (U.S. Department of Energy, January 16, 2025), 108, <https://www.energy.gov/eere/i2x/doe-distributed-energy-resource-interconnection-roadmap>.

conditioner cycling off, or clouds parting above a solar array) a spike in the net output power can be expected, and the PCS will not be able to respond instantly.

By default, UL 3141, the standard for PCS, requires that detecting and eliminating inadvertent export take 30 seconds or less in clause 8.2.3, while allowing utilities to require faster response times.⁶ While this response period is typically short enough to avoid equipment damage due to overloading (i.e. thermal damage to equipment), power quality impacts may be observed, particularly related to rapid voltage changes. To address this impact, a group of industry experts has proposed (and several states have adopted) a power quality screen which accounts for both the nameplate and export-limited capacity of proposed DER projects.⁷

The second caveat with export limited systems is noted in a footnote to the DOE report:

In general, PCSs are not considered under fault scenarios. Thus, screens for export-limited resources with PCSs would consider the nameplate rather than export capacity, and any impacts of inadvertent export should be captured in existing screens.⁸

That is, while export-limited capacity can and should be used when evaluating whether a given segment of the grid has capacity to accept a new generation resource, the full nameplate capacity of the resource must still be used when studying fault conditions to identify power quality impacts.

Notably, this caveat applies to more than just PCS-limited IBRs. No means of export limiting is instantaneous, regardless of the source of generation.⁹ Therefore, this also applies to

⁶ Underwriters Laboratories, “UL 3141 Outline of Investigation for Power Control Systems,” October 9, 2024, <https://www.shopulstandards.com/ProductDetail.aspx?UniqueKey=47260>.

⁷ IREC et al., *BATRIES: Toolkit and Guidance for the Interconnection of Energy Storage and Solar-Plus-Storage* (IREC, March 2022), 63–65, <https://energystorageinterconnection.org/wp-content/uploads/2022/03/BATRIES-TOOLKIT-FINAL-3.28.22.pdf>.

⁸ Baldwin et al., *Distributed Energy Resource Interconnection Roadmap; Transforming Interconnection by 2035*, 109.

⁹ PCS reply slower than traditional systems utilizing relays, which have been the subject of more recent and detailed discussion.

transmission-connected systems: the full nameplate capacity must be used when studying fault conditions.

Currently, we are not aware of studies or research on inadvertent export from transmission-connected generators. However, clause 4.6 of IEEE 2800-2022, the standard for such systems, does allow transmission system (“TS”) operators to specify external control inputs that IBRs must be capable of responding to, including “Capability to limit active power at a value below IBR continuous rating (ICR) within a time as specified by the TS operator.”¹⁰ (The forthcoming test and verification standard for IEEE 2800 will likely include more detailed criteria for verifying performance.¹¹) Notably, while allowing for this capability, IEEE 2800 makes no allowance for systems to be evaluated at anything less than nameplate capacity.

Despite the lack of information specifically related to inadvertent export on the transmission system, the available research for distribution-connected IBRs paints a compelling parallel: while so-called hybrid facilities certainly can connect to the transmission system, there are challenges that arise from the technical implementation of export limits that must be studied and understood, and the full nameplate capacity must still be used when studying fault conditions. We encourage further consideration of tools and techniques to address pragmatic protection requirements for protection from inadvertent exports or rapidly changing loads. The ANOPR includes additional expectations, such as in the following Principle 6. A future set of proposed rules forthcoming from the Commission should be informed by technical considerations and not be limited to a financial penalty framework.

¹⁰ The Institute of Electrical and Electronics Engineers, Inc., “IEEE Std 2800-2022: For Interconnection and Interoperability of Inverter-Based Resources (IBRs) Interconnecting with Associated Transmission Electric Power Systems” (IEEE, February 9, 2022), doi:10.1109/IEEESTD.2022.9762253.

¹¹ The Institute of Electrical and Electronics Engineers, Inc., “IEEE Std P2800.2: Draft Recommended Practice for Test and Verification Procedures for Inverter-Based Resources Interconnecting with Bulk Power Systems” (IEEE, October 23, 2025), <https://ieeexplore.ieee.org/document/11217395>.

B. Principle 6: System protection facilities for large loads with on-site, grid connected generation must include capacity, voltage, and duration limits

The sixth principle in the ANOPR seeks comment on operational limitations of hybrid facilities:

Sixth, any hybrid interconnection shall be required to install the system protection facilities necessary to prevent unauthorized injections or withdrawals that exceed the respective rights. We seek comment on whether other operational limitations should be considered. We also seek comment on the minimum technical requirements for such system protection facilities, whether a hybrid interconnection customer should be subject to penalties for unauthorized injections or withdrawals, how any such penalties should be designed, and how such penalties should be allocated to other transmission customers.¹²

We provide comment below on other operational limits and minimum technical requirements. We leave questions of penalties and allocation thereof to others.

Following and complimenting the discussion of inadvertent export above in discussion of Principle 5, operational limits for hybrid systems should include limits on duration of inadvertent export (or import) and the voltage change resulting from any inadvertent export or import (in addition to capacity limits). That is, system protection facilities must:

1. Ensure that if the facility exceeds the allowed limits, its import or export is returned to the allowed limits within a fixed time period, defined by the TSO.
2. During the inadvertent export or import event, maintain the voltage at the point of interconnection within limits defined by the TSO.

Regarding minimal technical requirements, while we cannot respond specifically, we recommend that the Commission convene a workgroup of technical experts to develop such

¹² Chris Wright, “Ensuring the Timely and Orderly Interconnection of Large Loads: Advance Notice of Proposed Rulemaking,” 12.

requirements. These requirements should be responsive to available technical capability and should prioritize reliability while avoiding excessive costs to developers and consumers.

C. Principle 7: Clear rules are required to enable this principle to be useable and non-discriminatory

The ANOPR set for consideration the following principle:

The interconnection study of large loads that agree to be curtailable and hybrid facilities that agree to be curtailable and dispatchable should be expedited. The system operator's ability to control such facilities through curtailment and/or dispatch must be sufficient for the system operator to integrate the facility into both operations and system planning.

The seventh principle recognizes the size and potential impact of many proposed large loads will require adjustments to conventional planning and operations to allow safe and reliable integration. To reflect this principle in a useful rule, the industry will look to the Commission for guidance on setting the basic terms of curtailment and/or dispatch. To implement this principle, the Commission should propose rules that both supply (in the case of a hybrid interconnection) and the load will need to commit to respond to operator dispatch instructions. That commitment should indicate both the duration and the quantity of the response, in either megawatts or percentage of the injection and withdrawal rights.

The Commission should also recognize that the large load will have an economic incentive to minimize the amount of capacity and the number of hours that are required to be curtailed and/or dispatched. Any forthcoming rule from the Commission must include criteria for determining the amount and the duration of curtailment and/or dispatch to avoid controversy and delay in the adoption of this principle. The most direct approach compatible with existing grid operator dispatch is to have both the supply and demand from such interconnection customers be designated as non-firm and subject to economic

dispatch. Existing practices and rules for both generation interconnection and types of transmission service recognize this level of service.¹³

In addition to the interconnection customer agreement to provide non-firm and dispatchable interconnection study and status, system operator tariffs must include provisions for planning and operation of such loads to be curtailable. Clear and simple designation of such resources throughout study, planning and operations protocols should be established for unambiguous and non-discriminatory application of this principle. Any rulemaking to establish this sort of treatment for large loads should anticipate or preferably include the reforms to planning and operations that will establish the full suite of practices needed to implement this principle in a nondiscriminatory manner.

D. Principle 8: Applying cost causation for interconnection costs will address a regulatory gap

The ANOPR set for consideration the following principle:

Load and hybrid facilities should be responsible for 100% of the network upgrades that they are assigned through the interconnection studies.

This principle should be adopted by the Commission in any future rulemaking regarding the interconnection of large loads. Without action by the Commission to address cost causation

¹³ See discussions of non-firm transmission service in the Order 888 adoption of the pro-forma Open Access Transmission Tariff. *Promoting Wholesale Competition Through Open Access Non-Discriminatory Transmission Servs. by Pub. Utils.; Recovery of Stranded Costs by Pub. Uti/so& Transmitting Uti/s.*, Order No. 888, FERC Stats. &Regs. ~ 31,036 (1996) (crossreferenced at 75 FERC ~ 61,080), *order on reh 'g*, Order No. 888-A, FERC Stats. & Regs. ~ 31,048 (cross-referenced at 78 FERC ~ 61,220), *order on reh 'g*, Order No. 888-B, 81 FERC ~ 61,248 (1997), *order on reh 'g*, Order No. 888-C, 82 FERC ~ 61,046 (1998), *aff'd in relevant part sub nom. Transmission Access Poly Study Grp. v. FERC*, 225 F.3d 667 (D.C. Cir. 2000), *aff'd sub nom. New York v. FERC*, 535 U.S. 1 (2002). For a broader context for customer service that is not assured for every hour of the year, see Alexandra B. Klass & Dave Owen, *Allocating Electricity*, 94 Geo. Wash. L. Rev. (forthcoming 2026), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5399325.

in transmission expansion for large loads, consumers will be burdened with billions of dollars in new costs added to rates. Costs for transmission connections and network extensions required by new large loads have already added over \$4 billion to the ratebase of utilities in 2024 from construction in just seven states.¹⁴

i. How large load connection costs are included in customer rates

The extraordinary size of individual new large load facilities generally obligate the use of higher voltage equipment to safely connect the loads to the grid. Most often this leads to large loads being directly connected to the transmission (high voltage), rather than the distribution (low voltage) system. As such, necessary system upgrades to connect these large loads also happen at the transmission rather than the distribution level. The cost-causation principle of allocating network upgrade costs breaks down, however, because current practices and protocols typically dictate that transmission costs are passed on to the load-serving entities as wholesale costs that are socialized across all retail customers in that utility's service territory.

The existing rules and practices allow the worst outcome: the rapidly rising costs for data centers' needs are spread across all the utilities' customers. Without reforms, existing rules will continue to allow the costs for connecting large data centers to the existing transmission system to raise electricity bills for all customers served by the same utilities.

ii. Transmission costs for large load customers fall through a regulatory gap

Utility retail rate filings are made at state utility commissions with the transmission costs set by FERC-formula rates. The annual revenue requirements for transmission are provided to

¹⁴ M. Jacobs. *Connection Costs*. Union of Concerned Scientists. September 29, 2025. <https://www.ucs.org/sites/default/files/2025-09/PJM%20Data%20Center%20Issue%20Brief%20-%20Sep%202025.pdf>

state regulators, but utilities do not typically distinguish or separate data center connection costs from other transmission costs spread across all the utilities' customers.

Transmission-level connections for a large load create costs that are often amalgamated with typical transmission costs that are driven by reliability needs, equipment replacements and collective customers' use and passed through to retail customers via state PUC approved rate cases. Any tracking of cost causation for the new large loads' transmission-level connection are obscured in the current process.

In a retail rate case, state regulators set different rates for classes of customers to reflect the costs to serve classes of similar customers and the revenues from those customers.¹⁵ Retail rate cases normally include an important step to do this: the Cost of Service Study, which sorts which customers rates should include which costs.¹⁶ With the lack of information in these proceedings about large customers causing transmission costs, there is little opportunity in the retail rate-setting process for costs to be assigned to a customer class of very large users, (i.e. users that connect at the transmission level).

To address the question of whether the transmission costs are somehow addressed by the revenues from new large load customers through existing ratemaking practices, the allocation of revenues is also critical to Principle 8. Without some reform, existing practices will continue where costs caused by large customers are generally only partially offset by revenues from the large customers. In a recent state inquiry regarding interconnection and tariffs of large loads, one Pennsylvania utility explained that in a retail rate-setting process, this incomplete information regarding costs prevents the regulator from offsetting the increased transmission costs with

¹⁵ Lazar, Jim, Paul Chernick, William Marcus, and Mark LeBel, eds. 2020. *Electric Cost Allocation for a New Era: A Manual*. Montpelier, VT: Regulatory Assistance Project. Pages 29-30.

¹⁶ Lazar, et al. at 15.

increased revenues from new large customers.¹⁷ Specifically, the revenues from a class of customers are collected and applied to meet the revenue requirements of that class of customers.¹⁸ Thus, the revenues from new large load customers are allocated only to the largest customer class and are not benefiting all customers.¹⁹ Under existing practices the revenues from new large load customers are recognized for the largest customer class, but the costs caused by those customers are allocated to all customers. This is not the application of the “Bonbright” cost-causation principle that is so often cited in utility cost allocation debates.²⁰

iii. Tracking transmission costs for large loads to rate base and revenue requirements

Utilities currently perform engineering studies for the interconnection of large loads in response to a request to the retail utility by the potential new customer. When the large size exceeds the capacity of existing distribution facilities, the study is performed by the Transmission Owner. In RTO regions, this customer-driven need is classified as a local project and, therefore, the costs of any required transmission upgrades are allocated to the local utility.

Transmission Owners in PJM provide enough information about their local planning²¹ that UCS was able to identify 150 examples of electric utilities connecting new customers with projected loads of 20 MW or greater at a cost to consumers of more than \$5 billion.²² These are costs the Transmission Owner identified through interconnection

¹⁷ Before The Pennsylvania Public Utility Commission En Banc Hearing on Interconnection and Tariffs for Large Load Customers. Docket No. M-2025-3054271. Comments of Duquesne Light Company. June 6, 2025. page 7. <https://www.puc.pa.gov/pdocs/1882372.pdf>

¹⁸ Lazar et al. at 29-30.

¹⁹ Before The Pennsylvania Public Utility Commission En Banc Hearing on Interconnection and Tariffs for Large Load Customers. Docket No. M-2025-3054271. Comments of Duquesne Light Company. June 6, 2025. page 7.

²⁰ James C. Bonbright. *Principles of Public Utility Rates*. Columbia University Press. New York. 1961.

²¹ PJM M-3 process described in PJM Interconnection *Operating Agreement*. Schedule 6 § 1.3.

²² M. Jacobs. September 2025.

studies as necessary for connecting the new customers, often including new transformers and substation gear on the property of the new customer. The information provided in the PJM process includes the need identified by the utility, and a tracking number that follows from the utility's locally planned transmission into the PJM Regional Transmission Expansion Plan ("RTEP").²³ Once included in the RTEP, that tracking number is then used to include the costs for these investments in the Annual Transmission Revenue Requirement for the calculation of the transmission component of retail rates.²⁴

In summary, this proposed Principle of assigning costs for all upgrades identified in the interconnection studies for large loads will address a regulatory gap and correctly apply the widely cited principle of cost causation to significant costs presently misallocated to all consumers. The ANOPR proposes a solution that encompasses network upgrades. By defining the problem to include network upgrades, the ANOPR addresses the particular problem of more extensive, and expensive transmission expansion frequently designed to connect large loads in recent years.

II. Conclusion

The Commission has initiated a rulemaking process that is both needed and broad. The challenges of rapid growth in demand from large loads are significant. The need to mobilize capital without discrimination or unjust rates remains pressing. We look forward to any Notice of Proposed Rulemaking that may follow the Commission consideration of this ANOPR.

²³ M. Jacobs. September 2025.

²⁴ See e.g. PJM Open Access Transmission Tariff Attachments H-1 through H-34.

Respectfully submitted on this 21st day of November 2025.

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